

GRAPHIC ERA HILL UNIVERSITY, DEHRADUN

SEMESTER I

Name of Department: - School of Computing

1.	Subject Code:	TBC 103	Course Title:	Mathematical foundation of Computer Science
2.	Contact Hours:	L: 3	T: 0	P: 0
3.	Examination Duration (Hrs):	Theory 3	Practical	0
4.	Relative Weight:	CIE 25	MSE 25	ESE 50
5.	Credits:	3		
6.	Semester:	I		
7.	Category of Course:	DSC		
8.	Pre-requisite:			

9. Course Outcome:	After completion of the course the students will be able to: CO1: Define and identify the basic concepts of set theory, relations and number theory. CO2: Interpret Venn-Euler diagrams, relations, functions, matrices and determinants. CO3: Use mathematical induction to prove mathematical statements. CO4: Analyze the properties of integers using the Euclidean algorithm and prime factorization. CO5: Analyze and solve complex real-life problem using concepts of probability theory.
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10. Details of the Course:

Sl. No.	Contents	Contact Hours
1	Unit 1: Set Theory: Introduction , Set and its Elements : Elements of a Set , Standard Sets and Symbols Set Description : Roster Method , Set Builder Method Cardinal Number (Dimension or Order) of a Set , Types of Sets , Venn-Euler Diagrams Set Operations and Laws of Set Theory : Union of Sets , Intersection of Sets , Disjoint Sets , Difference of Two Sets , Complement of a Set , Distributive Laws , Symmetric Difference of Sets Fundamental Products , Index and Indexed Sets , Partitions of Sets , Minsets, Countable and Uncountable Sets , Algebra of Sets and Duality , Computer Representation of Sets , The Inclusion and Exclusion Principle	9

2	Unit 2: Relations and Functions: Introduction Cartesian Product of Sets : The Cartesian Product of n Sets , Important Results on Cartesian Product Binary Relations : Binary Relation Defined in a Set , Domain and Range of a Relation Set Operations on Relations Types of Relations : Properties of Relations Partial Order Relations , Equivalence Relation , Functions: Introduction Definition and Notation of a Function : Range and Domain of a Function , Function as Sets of Ordered Pairs , Difference between Relation and Function , Difference between a Function and its Value Types of Functions , Invertible Functions Composition of Functions : Important Results on Composition of Functions , Identity Function Functions for Computer Science : Floor and Ceiling Functions , Fibonacci Sequence , Ackermann's Function , Characteristic Function, Mod Functions , Time-complexity Function	9
3	Unit 3: Number Theory: Introduction , Basic Properties of Integers , Properties of Integers , Division Theorem (or Algorithm) Greatest Common Divisor : Basic Properties of the Greatest Common Divisor Euclidean Algorithm: Basic Properties of Prime Factors Least Common Multiple , Testing for Prime Number	9
4	Unit 4: Mathematical Induction, Recursion and Fundamentals of Probability Theory Introduction , Principle of Mathematical Induction , Recursive Definitions and Structural Induction , Recursive Algorithms , Introduction to Probability Theory Concepts of Probability: Random Experiment , Sample Space , Types of Events	9
5	Unit 5: Determinants and Matrices Introduction to Determinant ,Determinant as Eliminant ,Minor , ,Cofactor , Rules of Sarrus , Properties of Determinants , Introduction to Matrices , Types of matrices , Addition of Matrices , Properties of Matrix Addition , Subtraction of Matrices , Scalar Multiple of a Matrix, Matrix Multiplication , Properties of Matrix Multiplication	9
	Total	45

Text Books:

Authors Name	Title	Edition	Publisher, Country	Year
J. K. Sharma	Discrete Mathematics	4 th Edition	Trinity Press	2005

H. K. Dass	Advanced Engineering Mathematics	21 st Edition	S.Chand and Company Ltd	2012
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Reference Books:

Authors Name	Title	Edition	Publisher, Country	Year
Kenneth. H. Rosen	Discrete Mathematics and its Applications	7 th Edition	McGraw Hill	2012
Gremaldi, Ramana	Discrete and Combinatorial Mathematics	5 th Edition	Pearson	2006